

Resilience Protocol: VR Stress Simulation

Presented by Ezra Shukurov



CSCI 5619-Virtual Reality and 3D Interaction

Motivation



Why VR + Stress?

- VR makes fear and stress feel *real* — impossible to replicate on a flat screen
- This realism enables **Stress Inoculation Training (SIT)** in a safe environment
- No weapons — player survives through **stealth, focus, and calm**
- Goal: build a **horror game that also functions as a training tool**

Related Work: Stress Inoculation in VR



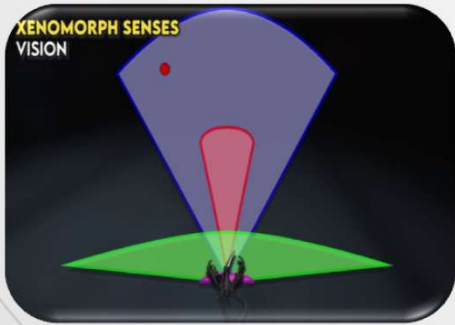
What is SIT?

- A method that exposes people to manageable, increasing levels of stress to build psychological resilience.
- VR induces authentic physiological responses (heart rate, adrenaline) that standard screens cannot replicate.
- The brain treats the virtual threat as real, making it a valid training environment.

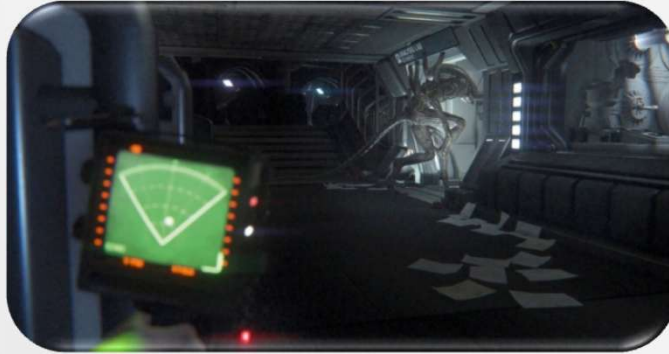
Real World Application

- I looked at a study involving students training for police and military service (Majewska & Zduńczyk, 2022).
- They measured "emotional states" using quantitative surveys. The VR group reported statistically significant increases in vigilance and fear, proving VR creates a unique "emotional training" environment that screens cannot match.
- Finseth et al. (2021) used biological monitoring during a VR space simulation to prove that "graduated stress" training builds significantly higher physiological resilience than skill-only practice.

Previous works



Alien: Isolation



Implementation



Fuse box and interaction
from *Resident Evil 7 VR*



Implementation

Metahuman enemy, sight
based dynamic chase

AI System

3D spatial enemy sounds
Puzzle sound effects

Audio



Interaction

Physics-based grabbing,
Fuse & keypad puzzles,
Hand-driven input

Gameplay

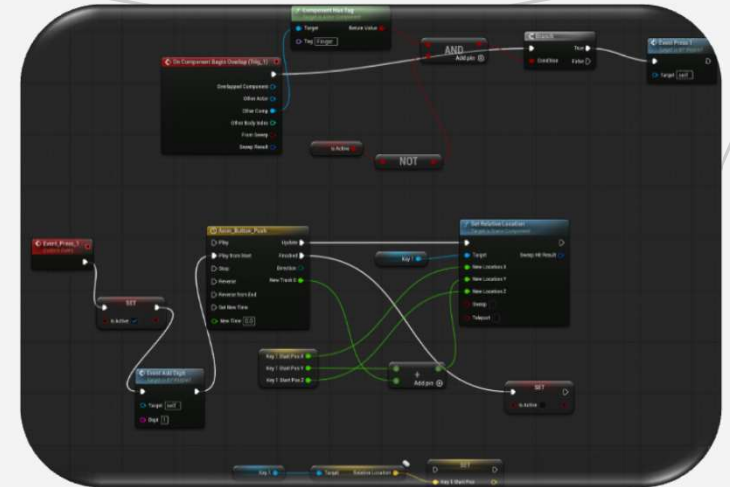
Linear escape flow,
Progressive difficulty,
No combat, stealth only

Optimization

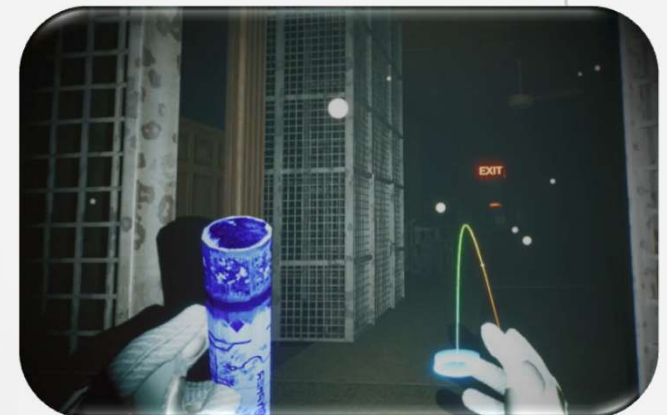
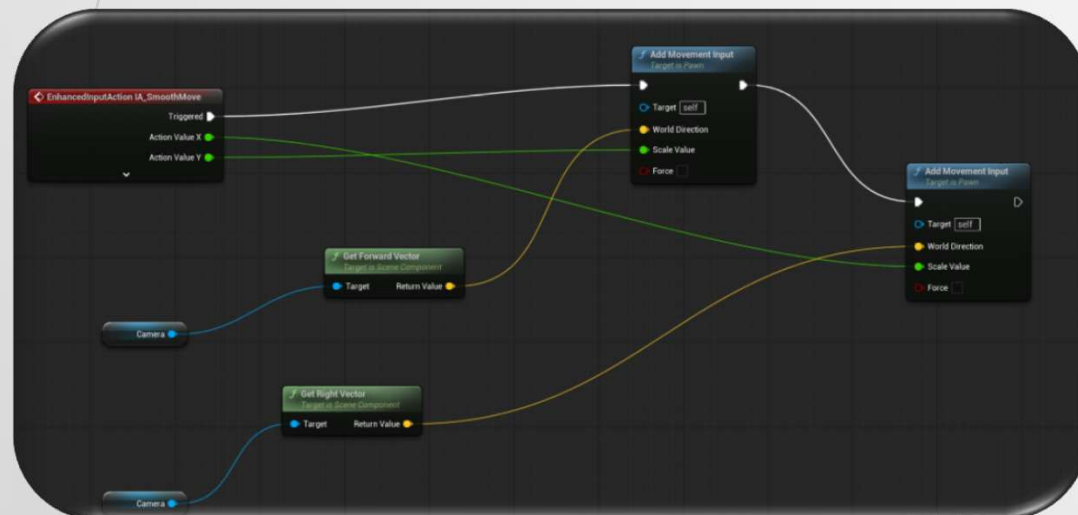
Forward rendering,
baked lighting,
Stable performance



Keypad logic



Locomotion methods



Assessment

Accomplished

- ✓ Physics-based puzzles
- ✓ Stable performance on Meta Quest 3
- ✓ Unpredictable AI behaviour
- ✓ 3D sound implementation

Issues and Future work

- ✗ Smarter AI
- ✗ Face animations for reality
- ✗ No real-time dynamic shadows
- ✗ No biometric stress measurement

Thank You!

